ENGINEERING THERMODYNAMICS (MME 365)

PERFECT (IDEAL) GASES

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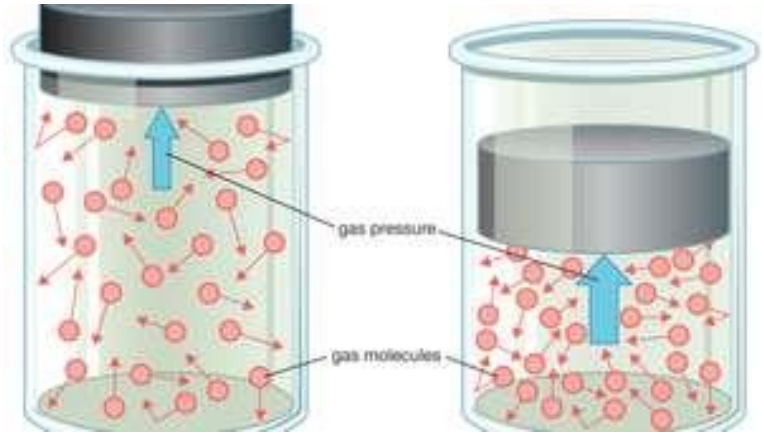
Perfect (Ideal) Gas

- One important fluid which has many applications in thermodynamics is the type in which the working temperature of the fluid remains well above the critical temperature of the fluid.
- In the simple treatment of such fluids, their behaviour is likened to that of a perfect gas.



Note

A perfect gas is an ideal gas which can never be realized in practice. However, the behaviour of many gases, e.g. hydrogen, oxygen, air, etc., is very similar to the behavior of a perfect gas to a first approximation.



Perfect (Ideal) Gas

- **Boyle's Law** :The volume of a given mass of gas is inversely proportional to the pressure of the gas provided the temperature remains constant.

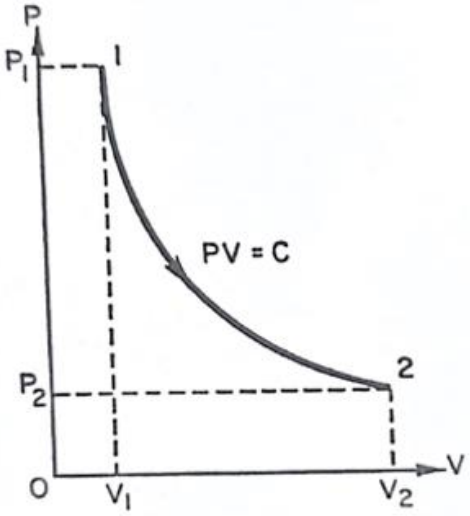
$$P \propto 1/V$$

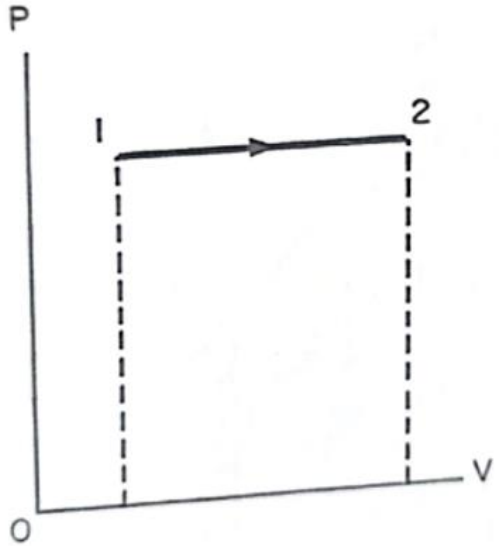
- **Charles' Law** :The volume of a given mass of gas is directly proportional to the temperature of the gas provided the pressure remains constant.

$$V \propto T$$

- **Combined Gas Law:** The volume of a given mass of gas is directly proportional to the temperature and inversely proportional to the pressure.

$$V \propto T/P$$





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Perfect (Ideal) Gas

The Characteristic Equation of State

- At temperatures that are considerably in excess of critical temperature of a fluid, and also at very low pressure, the vapour of fluid tends to obey the equation;

$$\frac{PV}{T} = k \text{ (constant)}$$

- If the mass of gas involved in the process is m kg, then dividing each side of the equation by m gives;

$$\frac{PV}{mT} = \frac{k}{m} = R$$

- R is called the specific gas constant (J/kg.K)

- The Characteristic equation of state of a perfect gas is normally written as

$$PV = mRT$$

- In terms of kilogram-mole, the equation of state of perfect gas can be written as;

$$PV = nR_uT$$

- R is the universal gas constant which has the value, 8314.3 J/kmol.K.
- If the molecular mass of a gas is known, the specific gas constant can be calculated using the expression;

$$R = \frac{R_u}{M}$$

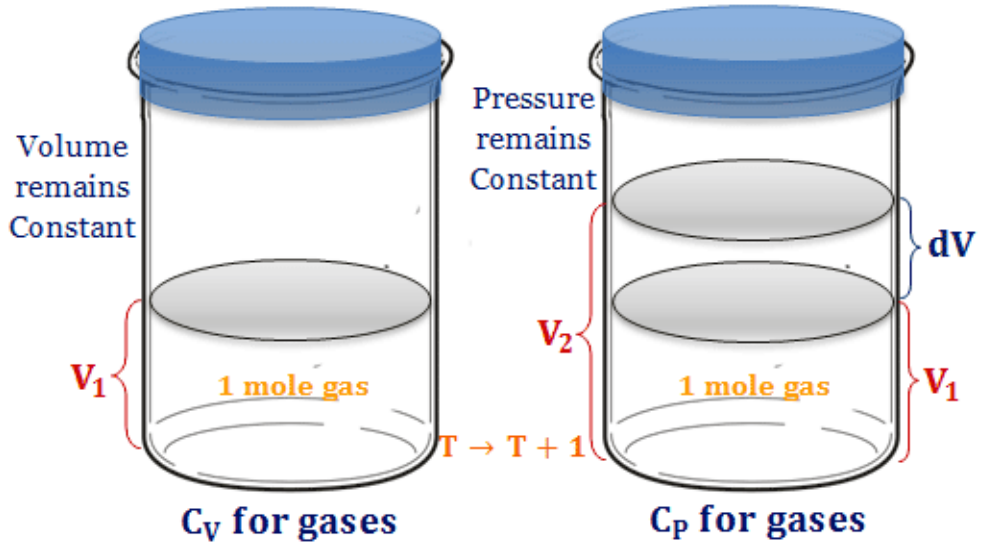
- M is the molecular mass in kg/kmol.

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Perfect (Ideal) Gas

Specific Heat Capacity

- It is defined as the amount of heat energy required to raise the temperature of unit mass of the substance through one degree, under specified conditions.
- In thermodynamics, two specified conditions are used, those of constant volume and constant pressure.
- The amount of heat energy supplied to 1 kg of a gas to raise the temperature by 1 °C while the volume remains constant, is known as the specific heat capacity at constant volume, and is denoted by c_v (J/kg.K).
- This is expressed as, $Q = mc_v\Delta T$
- The amount of heat energy supplied to 1 kg of a gas to raise the temperature by 1 °C while the pressure remains constant, is known as the specific heat capacity at constant pressure, and is denoted by c_p (J/kg.K).
- This is expressed as, $Q = mc_p\Delta T$



Perfect (Ideal) Gas

Joules' Law

- The internal energy of a perfect gas is a function of the absolute temperature only.

$$u = f(T)$$

- This implies that the internal energy varies linearly with absolute temperature. Internal energy can be made zero at any arbitrary reference temperature.
- For mass *m*, of a perfect gas, internal energy,

$$U = mc_v T$$

- For a perfect gas, in any process between states 1 and 2, change in internal energy,

$$\boldsymbol{\Delta U = U_2 - U_1 = mc_v(T_2 - T_1)}$$

Relationship Between Two Specific Heats

- Consider a perfect gas being heated at constant pressure from initial state 1 to final state 2
- According to the non-flow energy equation,

$$Q - W = \Delta U = mc_v(T_2 - T_1)$$

- In a constant pressure process,

$$W = P(V_2 - V_1) = mR(T_2 - T_1)$$
$$Q = m(h_2 - h_1) = mc_p(T_2 - T_1)$$

- Combining the equations above,

$$mc_p(T_2 - T_1) = mc_v(T_2 - T_1) + mR(T_2 - T_1)$$

- The relation between c_p and c_v is given by

$$\boldsymbol{c_p - c_v = R}$$

$$\boldsymbol{\gamma = \frac{c_p}{c_v} = \text{ratio of specific heats}}$$

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Non-flow Processes of a Perfect Gas

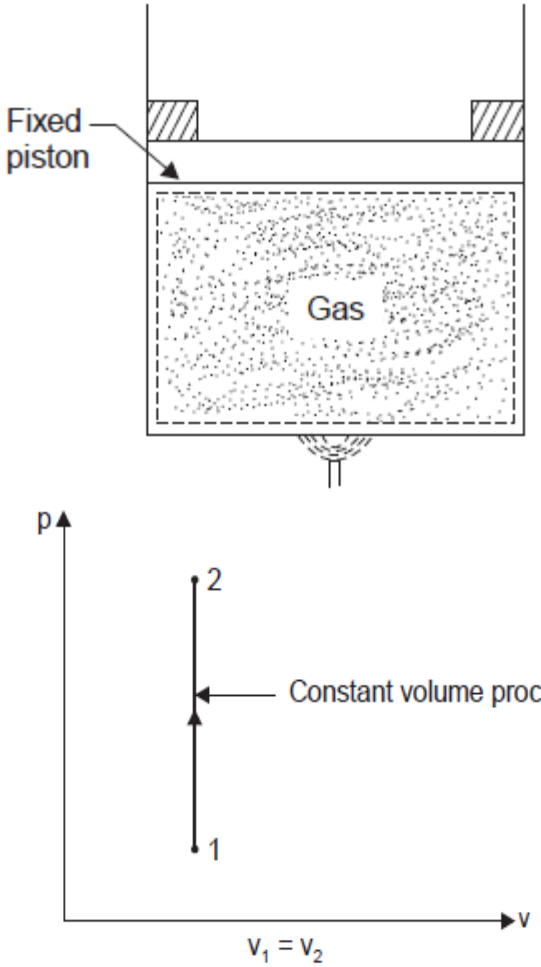
Reversible Constant Volume Process

- The working fluid is contained in a rigid vessel, hence the boundaries of the system are immovable and no work can be done on or by the system ($W_b = 0$), other than paddle-wheel work input.
- It will be assumed that '*constant volume*' implies zero work unless stated otherwise.
- Considering mass *m*, of the working fluid and applying first law of thermodynamics to the process;

$$Q - W = \Delta U = U_2 - U_1 \quad \text{but} \quad W = 0 \text{ (constant volume)}$$

$$\mathbf{Q = \Delta U = U_2 - U_1 = m(u_2 - u_1) = mc_v(T_2 - T_1)}$$

- **The heat supplied to a gas at constant volume is equal to the change in the internal energy of the gas.**



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Non-flow Processes of a Perfect Gas

Reversible Constant Pressure Process

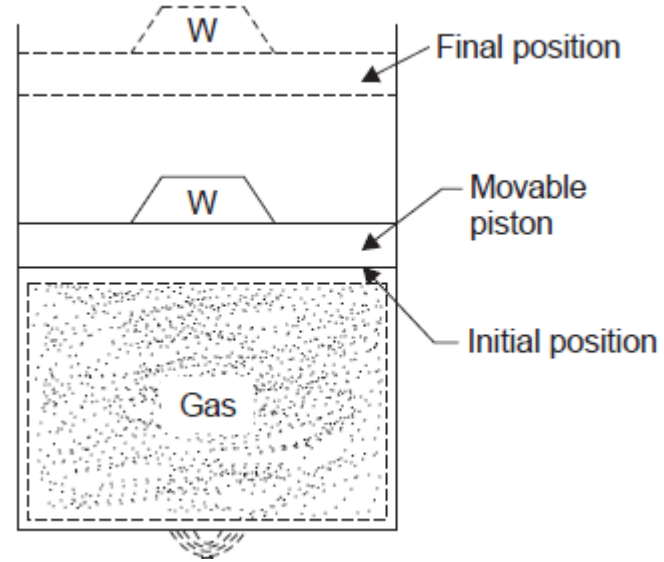
- For a constant pressure process, the boundary must move against an external resistance as heat is supplied.
- A gas in a cylinder behind a piston can be made to expand if the piston is pushed through a certain distance by the force exerted by the gas. Work is done by the gas on its surroundings.
- Considering mass *m*, of the working fluid and applying first law of thermodynamics to the process;

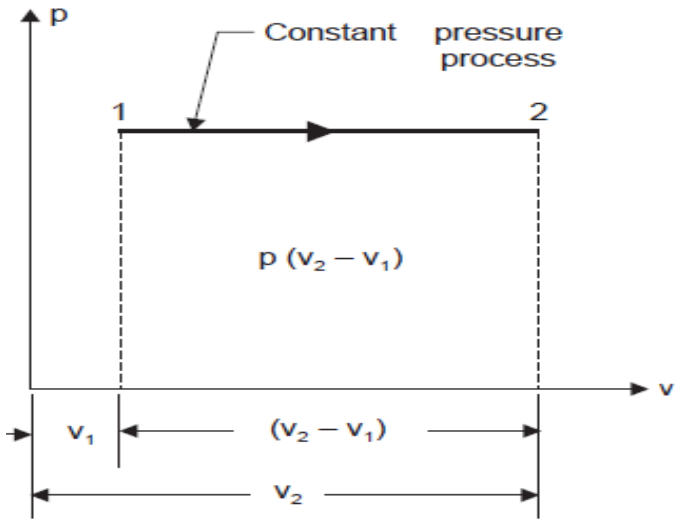
$$Q - W = \Delta U = U_2 - U_1 \quad \text{but} \quad W = P(V_2 - V_1) = mR(T_2 - T_1)$$

$$Q = \Delta U + W = mc_v(T_2 - T_1) + mR(T_2 - T_1)$$

$$\boldsymbol{Q = \Delta U + W = \Delta H = H_2 - H_1 = mc_p(T_2 - T_1)}$$

- **The heat supplied to a gas at constant pressure is equal to the change in the enthalpy of the gas.**

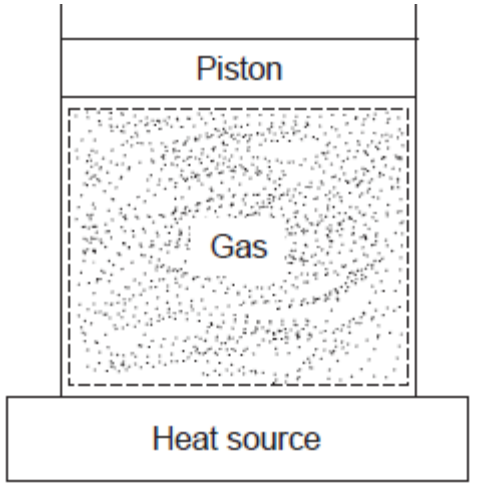




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Non-flow Processes of a Perfect Gas

Reversible Isothermal Process (PV = constant)

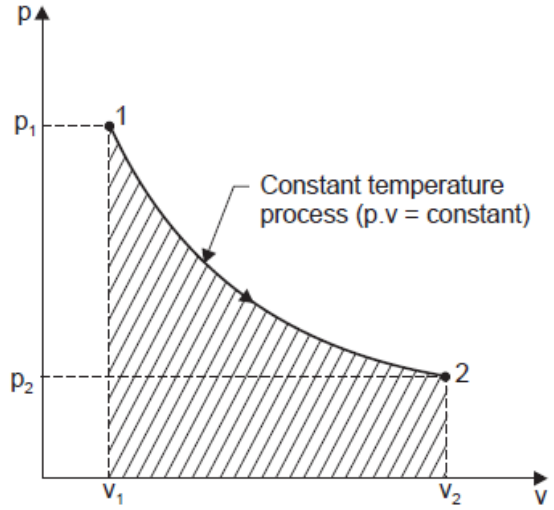


- When a working fluid in a cylinder behind a piston expands from a high pressure to a low pressure, there is a tendency for the temperature to fall.
- In an isothermal expansion, heat must be added continuously in order to keep the temperature at the initial value.
- Considering mass *m*, of the working fluid and applying first law of thermodynamics to the process;

$$Q - W = \Delta U = U_2 - U_1 \quad \text{but} \quad \Delta U = 0 \; (Const.temp)$$

$$Q = W = \int_1^2 p dV = \int_1^2 \frac{C}{V} dV$$

$$\boldsymbol{Q = W = P_1 V_1 ln \frac{V_2}{V_1} = P_2 V_2 ln \frac{V_2}{V_1} = mRT_1 ln \frac{V_2}{V_1}}$$

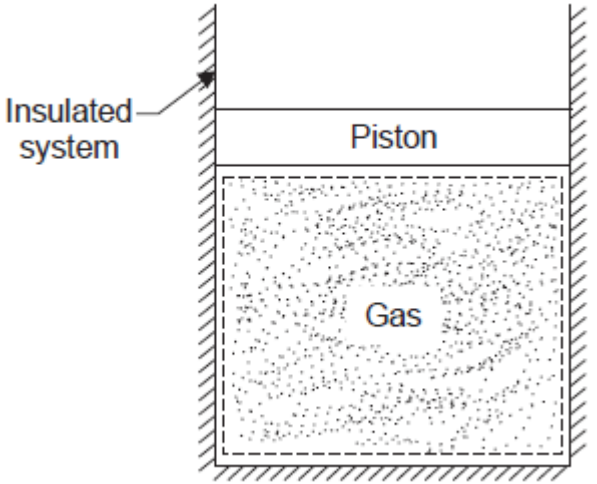


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Non-flow Processes of a Perfect Gas

Reversible Adiabatic Process (PV^{γ} = constant)

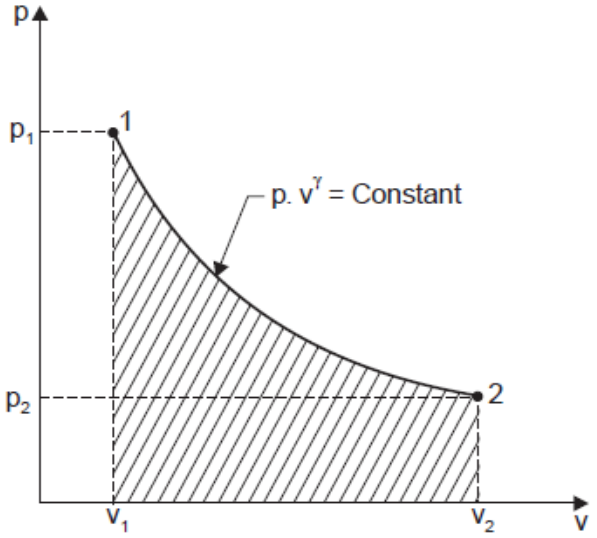
- An adiabatic process is one in which no heat is transferred to or from the fluid during the process. Such a process can be reversible or irreversible.
- The reversible adiabatic non-flow process will be considered.
- Considering mass *m*, of the working fluid and applying first law of thermodynamics to the process;



$$Q - W = \Delta U = U_2 - U_1 \quad \text{but} \quad Q = 0\ (adiabatic)$$

$$\Delta U = W = \int_1^2 p dV = \int_1^2 \frac{c}{V^{\gamma}} dV$$

$$\boldsymbol{\Delta U = W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{mR(T_1 - T_2)}{\gamma - 1}}$$



Non-flow Processes of a Perfect Gas

Reversible Adiabatic Process (PV^{γ} = constant)

❖ By using the equation of state PV = RT for a unit mass of gas, the relationship between T and V, T and P, and V and P may by derived using these two equations;

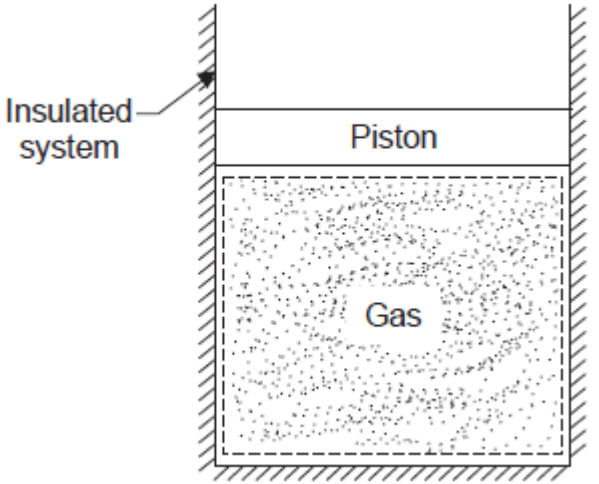
$$PV = RT \qquad (1)$$

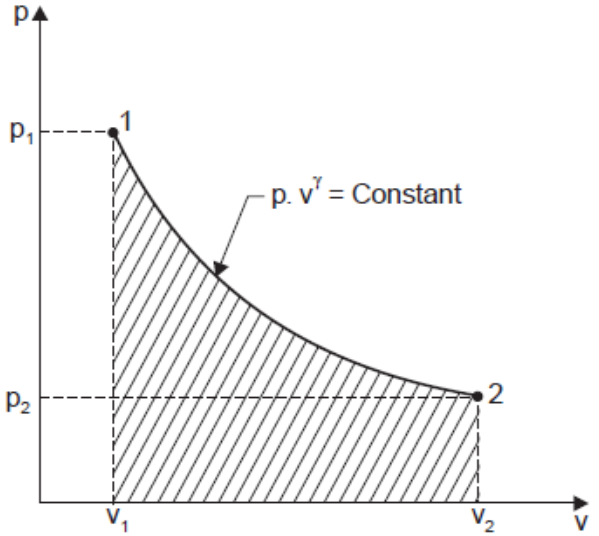
$$PV^{\gamma} = constant \qquad (2)$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} \qquad \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \qquad \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma}$$

$$\frac{\mathbf{P_2}}{\mathbf{P_1}} = \mathbf{r_p} = \textbf{pressure ratio}$$

$$\frac{\mathbf{V_1}}{\mathbf{V_2}} = \boldsymbol{r_v} = \textbf{compression ratio}$$





Summary of Processes for Perfect Gas (Unit mass)

Process	*Index* n	*Heat added*	$\int_1^2 pdv$	*p, v, T relations*	*Specific heat, c*
Constant pressure	$n = 0$	$c_p(T_2 - T_1)$	$p(v_2 - v_1)$	$\frac{T_2}{T_1} = \frac{v_2}{v_1}$	c_p
Constant volume	$n = \infty$	$c_v(T_2 - T_1)$	0	$\frac{T_1}{T_2} = \frac{p_1}{p_2}$	c_v
Constant temperature	$n = 1$	$p_1 v_1 \log_e \frac{v_2}{v_1}$	$p_1 v_1 \log_e \frac{v_2}{v_1}$	$p_1 v_1 = p_2 v_2$	∞
Reversible adiabatic	$n = \gamma$	0	$\frac{p_1 v_1 - p_2 v_2}{\gamma - 1}$	$p_1 v_1^{\gamma} = p_2 v_2^{\gamma}$ $\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma - 1}$ $= \left(\frac{p_2}{P_1}\right)^{\frac{\gamma - 1}{\gamma}}$	0

Perfect (Ideal) Gas Problems

Problem 1

0.06 m^3 of oxygen is contained in a sealed cylinder at a pressure of 1.38 bar and a temperature of 21°C. Determine the quantity of heat energy which must be supplied to the oxygen to increase its temperature to 93 °C, and the pressure of the oxygen at the final conditions.

The c_v and molar mass for oxygen are 0.986 kJ/kg.K and 32 kg/kmol, respectively.

Suggested Answers: [7.70 kJ; 1.72 bar]

Problem 2

A quantity of gas at 17.25 bar and 149 °C undergoes a non-flow constant pressure process which causes the volume of the gas to increase from 0.028 m^3 to 0.14 m^3. If $c_p = 1.005\ kJ/kg\ K$, and $c_v = 0.718\ kJ/kg\ K$, determine:

a)The change in internal energy

b)The work energy transferred

c)The heat energy transferred

Suggested Answers: [482 kJ; 194 kJ; 676 kJ]

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Perfect (Ideal) Gas Problems

Problem 3

A cylinder contains 0.28 m^3 of a perfect gas at 1,035 bar and 29 °C. The gas is compressed according to the law $PV^{1.3} = constant$ until the volume is reduced to 0.028 m^3. Heat energy is then supplied at constant pressure until the volume becomes 0.056 m^3 . Determine the temperature and pressure at the end of each process; the total change in internal energy; and the work energy transfer during each process. For the gas take cp and cv to be 1.06 kJ/kg.K and 0.75 kJ/kg.K .

Suggested Answers: [331°C, 20.7 bar; 935°C, 20.7 bar; 210.5 kJ; -96.7 kJ; 58 kJ]

Problem 4

A quantity of gas occupying 0.14 m^3 at 9.65 bar and 371°C is heated during a constant volume process until the pressure reaches 41.4 bar. The gas is then expand adiabatically to a pressure of 2.76 bar. Given that R=0.288 kJ/kg K and γ=1.41, calculate:

The temperature at the beginning of expansion,

The temperature at the end of expansion,

The heat energy supplied,

The work energy transferred,

Suggested Answers: [2487 °C; 982 °C; 108 kJ; 77 kJ]

Perfect (Ideal) Gas Problems

Problem 5

Gas at 1.035 bar and 21 °C with a specific volume 0.75 m^3/kg undergoes adiabatic compression through a volume ratio of 6:1. During the compression, the work energy transfer is 7.4 kJ, and the final temperature of the gas is 316 °C. Calculate the value of γ, R, c_p, c_v, and also the mass of gas involved in the process.

Suggested Answers: [1.387; 0.264 kJ/kg.K; 0.945 kJ/kg.K; 0.682 kJ/kg.K; 0.0368 kg]

Problem 6

0.14 m^3 of gas at 1.38 bar 38°C are compressed during a non-flow process according to the law $PV^{1.35}$ = constant, to a pressure of 20.7 bar. If R=0.264 kJ/kg K and γ=1.41, determine the change in internal energy, the work energy transfer, and the heat energy transfer.

Suggested Answers: [48 kJ; -56 kJ; -8 kJ]


THANK
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